

Homework 2

MATH 220

Problem 1 (Examples of Topologies)

- i) Prove that $\mathcal{T} = \{] - \infty, x[\mid x \in \mathbb{R} \} \cup \{ \emptyset, \mathbb{R} \}$ is a topology on $\mathbb{R}$.
- ii) Prove that $\mathcal{T} = \{ O_n \mid n \in \mathbb{N} \} \cup \{ \emptyset, \mathbb{N} \}$ where $O_n = \{ k \in \mathbb{N} \mid k \geq n \}$ is a topology on $\mathbb{N}$.

Problem 2 (Intersections and Unions of Topologies)

- i) Let $\{ \mathcal{T}_\alpha \}_{\alpha \in \mathcal{A}}$ be a family of topologies on a nonempty set X . Prove that $\bigcap_{\alpha \in \mathcal{A}} \mathcal{T}_\alpha$ is a topology on X .
- ii) Is it true that the union of two topologies on X is a topology on X ? Give a proof or a counterexample, as appropriate.

Problem 3 (Hausdorff condition and Metrizability)

A topological space X is **Hausdorff** if for all $x, y \in X$, $x \neq y$, there exist open sets U and V such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$. (This means that any two distinct points are contained in two disjoint open sets)

- i) Prove that, if a topological space is metrizable, then it is Hausdorff.
- ii) Prove that an indiscrete topological space that contains at least two points is not metrizable.
- iii) Prove that an infinite set endowed with the cofinite topology is not metrizable.